Condition-Dependent Cognitive Processing: Exploring Sleep-Related Factors, Emotional Regulation and Cognitive Performance Using Secondary Data

1st Independent Secondary Data Analysis Project

On GitHub

ABSTRACT:
Cognitive performance can vary as a function of both prior experience and current psychophysiological states. Using a publicly available secondary dataset, this project examined how experience-related and psychophysiological-state contributors were associated with emotion regulation and cognitive-task performance. An initial multivariate exploratory analysis characterized zero-order associations among the available behavioral, psychological, and cognitive variables. Aim 1 then used regression-based models to examine the independent and conditional associations of sleep duration and stress level with emotion-regulation performance, including a sleep-duration * stress-level interaction. Aim 2 first applied exploratory principal component analysis (PCA) to determine whether Stroop reaction time, N-back accuracy, and psychomotor vigilance test reaction time could be meaningfully summarized by a broader common cognitive-performance dimension. Because a dominant single-component solution was not supported, the cognitive measures were subsequently examined separately using regression, ANOVA, and ANCOVA models. Across the present sample, no examined zero-order correlations reached statistical significance, and Aim 1 provided insufficient evidence for independent or interactive associations of sleep duration and stress with emotion regulation. PCA indicated substantial task-specific variance rather than a dominant common component, and the subsequent task-specific analyses likewise provided limited evidence for the hypothesized associations. Overall, the findings highlight substantial heterogeneity across cognitive measures and suggest that the theoretically expected relationships were not clearly detectable in the present dataset.

INTRODUCTION:

Cognitive processing is one of the most fundamental topics in psychology and neuroscience, as it provides a framework for understanding functionally segregated yet integrated information-processing mechanisms underlying how individuals interact with the surrounding environment (Frank & Badre, 2015; Schall, 2004; Forstmann et al., 2011). Presented information may influence observable performance both directly, as the immediate input to cognitive processing, and indirectly, by interacting with the broader processing context under which cognition operates. For the present framework, this context includes both prior experience and the individual's current internal state.

Certain studies support a broader conceptualization of cognitive functioning as dynamically state- and context-dependent, such that processing outcomes may vary across contextual conditions even when the immediate processing input remains unchanged (Flavell et al., 2022; Gilbert & Sigman, 2007). For the present analyses, these contextual contributors are broadly distinguished into two categories. Experience-related contributors refer to influences arising from prior experiences or interactions with the surrounding environment that subsequently shape cognitive functioning. Examples include insufficient sleep and body posture (Alhola & Polo-Kantola, 2007; Khan & Al-Jahdali, 2023; Wainio-Theberge & Armony, 2025). In contrast, psychophysiological-state contributors refer to conditions manifested within the individual at psychological and/or physiological levels, such as emotional state, fatigue, or motivation (Flavell et al., 2022). For example, perception of emotional information may be biased by the perceiver's current emotional state (Trilla et al., 2021; Harris et al., 2016). These contextual influences may operate partly through modulation of neural and physiological mechanisms underlying cognitive functioning. Importantly, the two categories are not necessarily independent: psychophysiological states may be elicited or modified by prior experience, while relatively persistent conditions such as trait anxiety or depressive states may continue to influence cognitive performance even without an immediate external trigger (Pacheco-Unguetti et al., 2010; Roca et al., 2015). Conversely, internal states may also shape subsequent interactions with the surrounding environment.

The publicly available Kaggle dataset used in the present project provides a broad range of behavioral, psychological, and cognitive-performance measures (Sacramento Technology, 2024), allowing two broad purposes to be addressed. First, the overall association structure among the available variables was explored using a correlation matrix heatmap to characterize their zero-order relationships. Second, theoretically relevant contextual predictors were selected for more focused model-based analyses.

For the second purpose, sleep hours and stress-level scores were selected as representative experience-related and psychophysiological-state predictors, respectively. Previous studies have linked both factors to variation in cognitive functioning and performance across different cognitive tasks (Alhola & Polo-Kantola, 2007; Khan & Al-Jahdali, 2023; Nwikwe, 2025). The first specific aim was therefore to examine their independent contributions and combined explanatory value in relation to higher-order regulatory functioning. Emotion regulation was selected as the representative higher-order outcome because successful regulation relies on the coordination of cognitive processes including selective attention, working memory, and inhibitory control (Koay & Meter, 2023).

The dataset also contains task-based measures of more basic cognitive functions measured using Stroop task reaction time (RT), N-back accuracy, and psychomotor vigilance test (PVT) RT. This motivated the second aim: to examine whether these cognitive-performance measures could be reasonably summarized into a broader common dimension while retaining meaningful information from the original measures. If such a dimension could not be supported, the individual cognitive outcomes would instead be retained and examined separately. Subsequent model-based analyses were then used to examine theoretically relevant associations with these cognitive outcomes.

In summary, the present secondary data analysis project combined exploratory association mapping with theory-driven model-based analyses to evaluate how experience-related and psychophysiological-state factors were associated with emotion regulation and cognitive performance. No specific pairwise association in the exploratory analysis was treated as a confirmatory hypothesis; rather, the correlation structure was examined to characterize the overall pattern of relationships in the dataset. For the theory-driven analyses, the selected contextual predictors were expected to show meaningful associations with their corresponding emotional and cognitive outcomes, while the available cognitive measures were also examined for the possibility of a broader common performance dimension.

-Aim1 Overview and Methods

Aim 1 examined the contributions of experience-related and psychophysiological-state factors to emotion regulation. Emotion regulation is conceptualized as a higher-order regulatory function relying on coordinated executive processes (Schulze et al., 2026). Evidence suggests that concurrent cognitive demands can interfere with regulation success or cognitive-task performance, consistent with competition for cognitive-control resources (Friese et al., 2013; Gan et al., 2017). Neuroimaging studies further indicate recruitment of prefrontal control regions overlapping with systems supporting attention and inhibitory control (Kohn et al., 2014; Martin & Ochsner, 2016). Sleep duration and stress level were therefore selected as theoretically relevant predictors. Sleep deprivation has been associated with poorer adaptive emotion regulation and altered prefrontal–limbic control, supporting sleep duration as an experience-related predictor (Tempesta et al., 2018; Tomaso et al., 2021), whereas stress can alter prefrontal executive-control functioning and emotion-regulation effectiveness, supporting its treatment as a psychophysiological-state predictor (Tsai et al., 2019; Langer et al., 2025).

The Aim 1 analysis therefore proceeded sequentially. A simple linear regression first examined the association between sleep duration and emotion regulation, and longer sleep was hypothesized to predict higher emotion regulation scores as a positive association was expected. Stress level was then added in a multiple regression to estimate the unique conditional association of each predictor while adjusting for the other and to test whether stress explained additional variance beyond sleep hours alone. Eventually, a sleep-hours × stress-level interaction was introduced to examine whether the association between either predictor and emotion regulation depended on the level of the other. I tentatively expected a positive interaction, such that the negative association between stress and emotion regulation would be weakened as sleep duration increased. Equivalently, the beneficial effect that one extra hour of sleep had on emotion regulation would become stronger at higher stress levels.

-Aim2 Overview and Methods

Aim 2 further examined associations between experience-related and psychophysiological-state contributors and cognitive-task performance. In contrast to Aim 1, which focused on emotion regulation as a higher-order regulatory process, Aim 2 targeted more basic executive and attentional functions. Stroop RT, N-back accuracy, and PVT RT were treated as measures primarily reflecting selective attention and inhibitory control, working memory, and sustained attention and behavioral alertness, respectively (Basner et al., 2021; Owen et al., 2005; Scarpina & Tagini, 2017).

An exploratory PCA was first conducted to examine whether the three task measures could be summarized by a broader common performance dimension. If such a dimension were present, the first principal component was expected to account for a substantial proportion of the total variance and show a theoretically coherent loading pattern, with Stroop RT and PVT RT loading in the opposite direction to N-back accuracy because better performance is reflected by lower reaction times but higher accuracy. If a meaningful single-component solution was not supported, separate task-specific models were examined. For Stroop performance, a negative association between emotion-regulation scores and RT was tentatively expected. Because effective emotion regulation can involve attentional deployment (Wadlinger & Isaacowitz, 2011), stronger emotion-regulatory ability may also be associated with more efficient non-emotional attentional control if these mechanisms are partly domain-general. For N-back performance, higher stress was expected to be associated with lower accuracy, consistent with evidence linking stress to impaired working-memory and prefrontal executive functioning (Fourteen & Fabre, 2026; Schoofs et al., 2008). Finally, PVT performance was examined across low-, moderate-, and high-stress groups using ANCOVA, with emotion-regulation scores included as a covariate, to test whether sustained attention differed across broader stress levels conditionally after adjusting for individual differences in emotion regulation.

RESULTS AND DISCUSSION

Exploratory Data Analysis (EDA):

This project used the Dataset.csv file stored in the Report/Original_Dataset directory of the project repository. Prior to formal analyses, an initial descriptive and exploratory analysis was conducted. The dataset contained 60 observations and 14 variables. For these variables, only the variable gender is a binary categorical variable (male vs. female), while all other variables are continuous. Descriptive statistics were subsequently examined to characterize the distributions and central tendencies of the variables in the dataset. Here, Table 1 presents the descriptive summary of the primary continuous variables. Eventually, initial screening indicated that there were no missing or duplicate values in this dataset, as described; therefore, there was no need for imputation or further data cleaning.

Table 1.

Descriptive statistics for the primary continuous variables.

Variable	Min	Q1	Median	Mean	Q3	Max
Sleep Hours	3.1	4.1	5.7	5.8	7.3	8.8
Sleep Quality	0.0	4.0	8.0	8.3	13.0	20.0
Daytime Sleepiness	0.0	6.0	11.5	12.0	19.0	24.0
Stroop Task Reaction Time	1.6	2.6	3.3	3.2	4.0	4.5
N-Back Accuracy	50.9	64.6	74.3	75.0	85.6	99.7
Emotion Regulation Score	10.0	25.0	37.0	38.2	54.3	67.0
PVT Reaction Time	201.6	257.5	327.2	332.5	402.8	494.6
Age	18.0	21.8	28.5	29.5	36.0	43.0
BMI	18.7	23.6	27.4	27.3	30.8	35.0
Caffeine Intake	0.0	1.0	2.5	2.4	4.0	5.0
Physical Activity Level	0.0	1.0	4.0	4.1	6.0	10.0
Stress Level	0.0	8.8	17.5	17.9	26.3	40.0

***Note*.** Q1 and Q3 represent the first and third quartiles (25th percentile and 75th percentile).

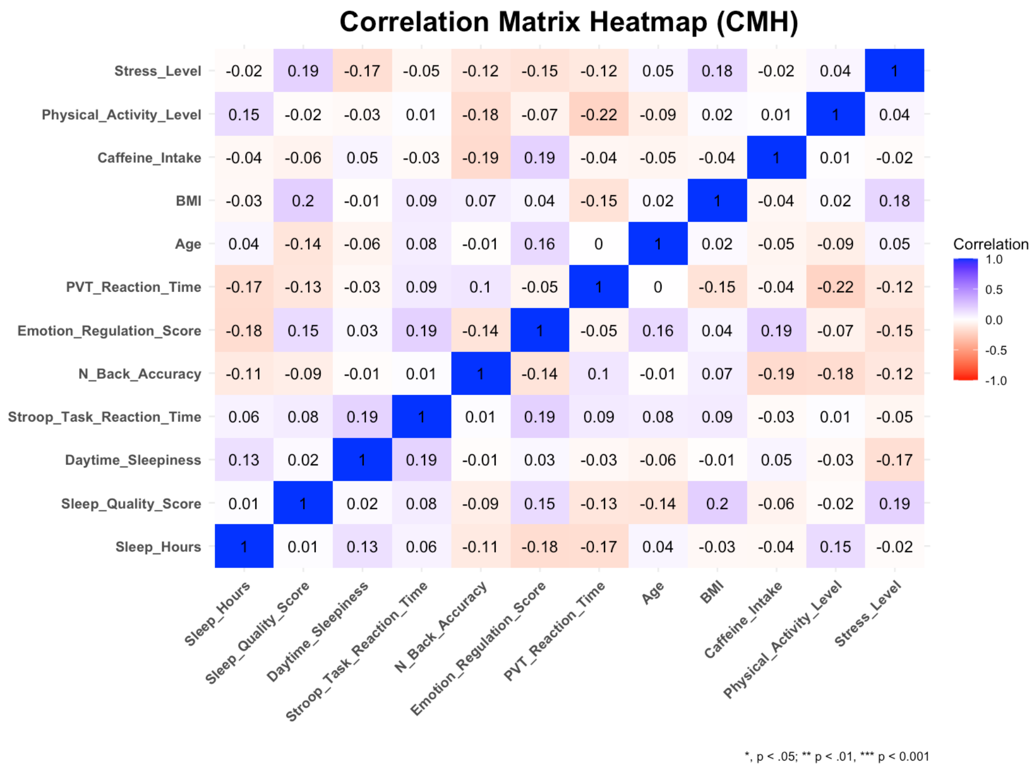
Table 1 also indicates that for continuous variables, the means and medians are closely aligned, and the distances from the median to the upper and lower quartiles are roughly equivalent. This observation reveals that the marginal distributions of outcome and predictor variables are approximately symmetric. The evidence thus preliminarily supports the use of linear regression as the modeling framework, obviating the need for variable transformations. Further diagnostic procedures based on model residuals will be performed to evaluate the validity of the inferential assumptions in formal analyses.

The Correlation Matrix Heatmap (CMH), showing zero-order correlations among variables, is useful in EDA because it provides a matrix that can simultaneously display the direction and magnitude of pairwise correlation estimates among multiple variables. The CMH was therefore applied beforehand for multivariate EDA, as shown in Figure 1.2. Visually, the CMH indicates that the magnitude of the observed standardized correlations was generally small and that none of them were statistically significant, suggesting that, in the present sample, there was insufficient evidence to support non-zero population-level pairwise correlations among the examined variables. Indirectly, this may also challenge some previous well-known established theories stating the effects of experience-related or psychophysiology-related factors on behavior, though these theories were not formally evaluated. For instance, scholars have suggested that sleep hours, as an experience-related factor, are negatively associated with waking behavior by manifesting as daytime sleepiness (Liu, et al., 2000; Oliveria, et al., 2025), but here, there is insufficient evidence for a meaningful correlation between the two variables.

Although the heatmap provides some basic information about inter-variable correlation, diagnostic assessment remains necessary because the validity of correlation-based inference depends on relevant statistical assumptions, and the absence of statistically significant zero-order correlations does not preclude the usefulness of further model-based analyses examining conditional associations and more specific theoretical relationships in subsequent formal analyses.

Figure 1.2

Correlation Matrix Heatmap (CMH) for Multivariate EDA



Note. Cell values represent zero-order Pearson correlation coefficients. Positive correlations are shown in blue and negative correlations in red, with color intensity indicating the magnitude of the correlation. No off-diagonal correlations reach statistical significance in the present sample.

-Analysis-Specific Data Preprocessing and Preparation

To provide a more practically meaningful interpretation of regression coefficients (e.g., values of 0 for sleep hours, emotion-regulation scores, and stress level are meaningless and unrealistic), mean-centering was conducted for every predictor variable of interest by shifting the reference point from the original value of 0 to the corresponding sample means. Table 2 shows the descriptive summary of the mean-centered predictor variables and the outcome variables of interest. Emotion regulation scores were retained in both their original and mean-centered forms. As mathematically expected, the means of all mean-centered predictor variables were approximately 0, while the variance and standard deviation of the mean-centered emotion regulation scores were the same as those of the original scores.

The PSS-10 provides a basis for transforming continuous stress-level scores into a categorical variable. A commonly used PSS-10 categorization criterion which has been validated in previous populations was applied to stress-level scores in this module for the ANCOVA in Aim 2, where stress group was treated as the factor. Basically, if an individual's stress-level score was less than 14, the individual was classified into the low-stress group; otherwise, if the score was between 14 and 26, the individual was classified into the moderate-stress group; otherwise, the individual was classified into the high-stress group (Adamson et al., 2020).

Table 2

Descriptive Statistics of Mean-Centered Predictors and Outcomes

Variable	Mean	Variance	SD
C_Sleep Hours	0.0	3.4	1.8
C_Emotion Regulation Scores	0.0	293.6	17.1
C_Stress_Level Scores	0.0	121.3	11.0
Emotion Regulation Scores	38.2	293.6	17.1
Stroop Task Reaction Time	3.2	0.7	0.8
N-Back Accuracy	75.0	186.9	13.7
PVT Reaction Time	332.5	7708.9	87.8

Note. C_Sleep hours, C_emotion regulation scores, and c_stress level scores represent mean-centered predictors, respectively.

-Construction of Functions for Assumption Diagnostics

Conceptually and mathematically, since the OLS coefficient estimator is a linear function of population errors conditional on the predictor structure, its corresponding sampling distribution depends on the distributional and variance properties of those errors. Thus, assumptions targeting the population error structure, assessed indirectly through residuals, should be reasonably satisfied to support accurate estimation of uncertainty and valid inference based on the theoretical sampling distribution. Here, to facilitate this, I implemented diagnostic procedures as reusable functions in a separate module, allowing consistent application across fitted models (see the "Symbols" section in the code file, where the defined assumption-diagnostic functions are listed).

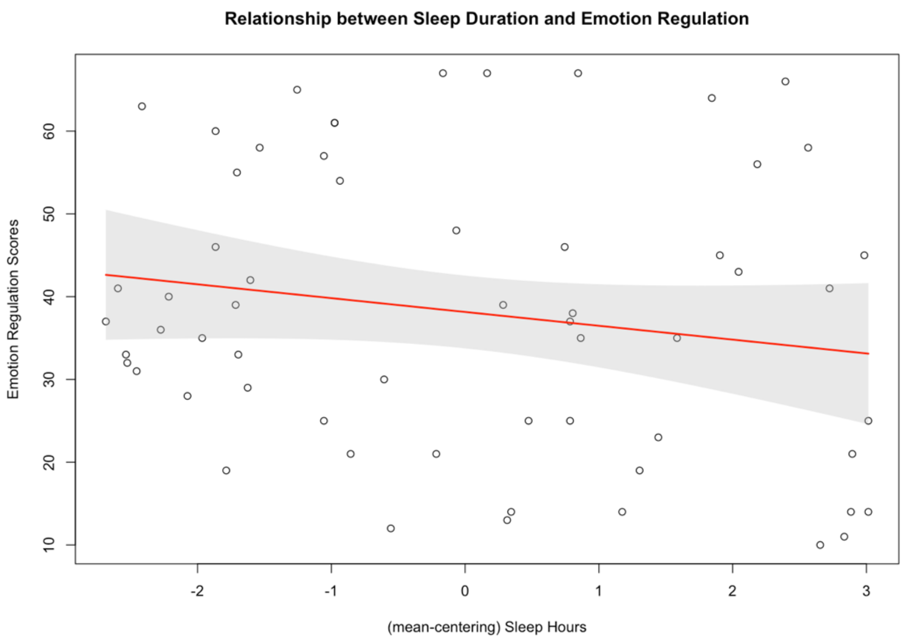
-AIM 1 Results and Discussion

Aim1_model1 established a simple linear regression to examine the effect of mean-centered sleep hours on emotion-regulation scores. At the sample level, the result indicates that each one-hour increase in sleep duration above the sample mean was associated with a 1.67-point decrease in emotion regulation score, but this result was not statistically significant, b = -1.67, SE = 1.21, t-value = -1.38, and p-value = 0.17 > 0.05. The model merely captured approximately 3.20% of the variance in emotion-regulation scores, R^2 = 0.03, apparently leaving most of the outcome variance unexplained by this model. Figure 2.1 shows the relationship between sleep duration and emotion regulation with a confidence interval band.

Assumption diagnostics were then conducted to assess whether the model conditions supporting valid statistical inference were reasonably satisfied. Pre-written diagnostic functions were applied sequentially, except for multicollinearity diagnostics, which were not required for this single-predictor model. For normality, the Q-Q plot and Shapiro-Wilk (S-W) test provided visual and statistical evidence of deviation from normality, p=.013 < 0.05 in the S-W test. Homoscedasticity, independence of residuals, linearity, and outliers were checked, and no substantial violations were detected (for diagnostics on other assumptions, see "Figure" section in the report file). To reduce reliance on the normality assumption for inference, a nonparametric bootstrap was used to approximate the sampling distribution of the coefficient. The bootstrapped 95% confidence interval (5000 resamples) was [-3.96, 0.54], including 0. Therefore, the bootstrap analysis likewise provided insufficient evidence that sleep duration was not associated with emotion-regulation scores at the population level.

Figure 2.1

Relationship between Mean-Centered Sleep Hours and Emotion Regulation Scores



Note. The shaded area represents the 95% confidence interval around the fitted regression line for the estimated mean outcome at corresponding values of the predictor.

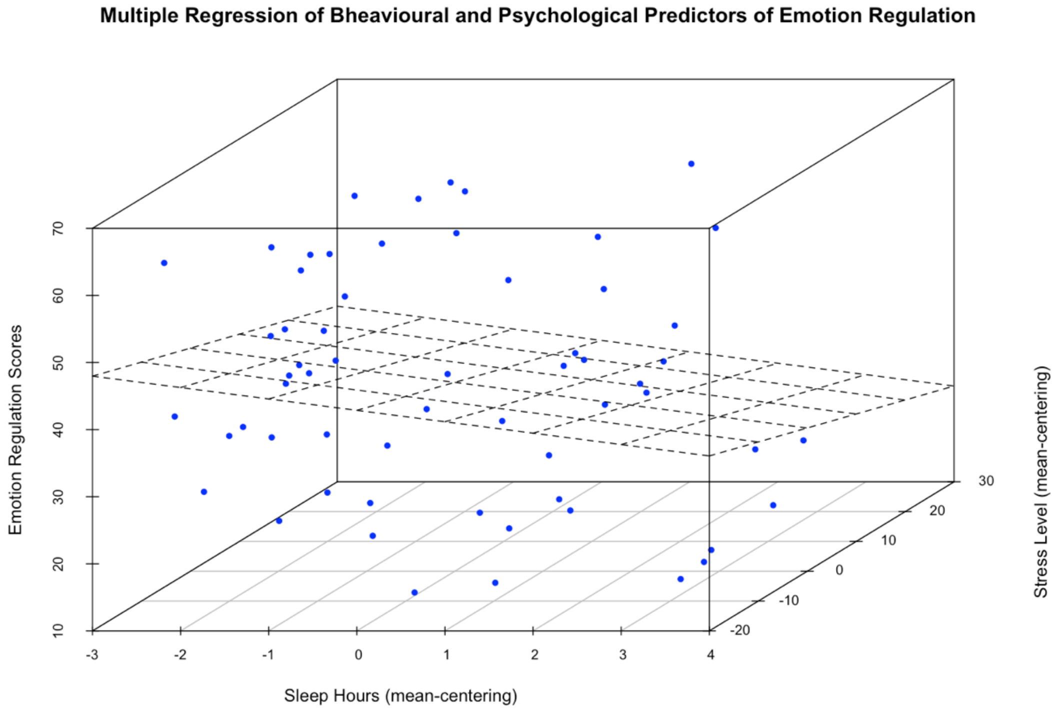
Aim1_model2 established a multiple linear regression by incorporating mean-centered stress-level scores as a new predictor variable. The effects of sleep hours and stress levels were now conditional. At the sample level, the result shows that each one-hour increase in sleep duration above the sample mean was associated with a 1.70-point decrease in emotion regulation score when stress level was controlled at its mean. A one-point increase in stress level above the sample mean was associated with a 0.24-point decrease in emotion-regulation score when sleep duration was controlled at its mean. However, neither of these results was statistically significant. For the conditional effect of sleep duration, b = -1.70, SE = 1.20, t-value = -1.41, and p-value = 0.16 > 0.05. For the conditional effect of stress level, b = -0.24, SE = 0.20, t-value = -1.18, and p-value = 0.24 > 0.05. The model accounted for approximately 5.5% of the variance in emotion-regulation scores, R^2 = 0.055, with an adjusted R^2 = 0.022, apparently leaving most of the outcome variance unexplained by this model, with the incorporation of the new predictor variable. Figure 2.2 shows the respective conditional effects of the predictor variables and the hyperplane representing the model fit that they construct in 3D space.

Regarding normality, the Q-Q plot and Shapiro-Wilk (S-W) test provided visual and statistical evidence of deviation from normality, p-value =.04 < 0.05 in the S-W test. Homoscedasticity, independence of residuals, linearity, outliers, and multicollinearity were checked separately, and no substantial violations were detected (for diagnostics on other assumptions, see “Figure” section in the report file). To reduce reliance on the normality assumption for inference, a nonparametric bootstrap was used to approximate the sampling distribution of the coefficient. The bootstrapped 95% confidence intervals (5000 resamples) for sleep duration and stress level were [-4.11, 0.71] and [-0.64, 0.17], respectively, and both included 0. Therefore, the bootstrap analysis likewise provided insufficient evidence that sleep duration was associated with emotion-regulation scores at the population level. However, for stress level, it was observed that its bootstrapped 95% CI was relatively narrower than that for sleep hours. This may possibly indicate that relatively, there was less uncertainty in the sample statistic for the regression coefficient of stress level. This model was subsequently compared with Aim1_model1 to evaluate whether the addition of stress level improved explanatory performance beyond sleep duration alone. The addition of stress increased R^2 by approximately 0.023, while adjusted R^2 increased by approximately 0.007. The nested-model comparison was not statistically significant, p= 0.24 < 0.05, providing insufficient evidence that stress level contributed incremental explanatory value beyond the sleep-duration-only model.

Following Aim1_model2, moderation_model3 was constructed by incorporating an interaction term (mean-centered sleep duration * mean-centered stress level). Whereas the additive model assumed that the association of each predictor with emotion regulation remained constant across values of the other predictor, the interaction model examined whether these associations varied conditionally across predictor levels. It also sought to test whether the non-significant average associations observed in the additive model might have been obscured by effects that differed across levels of the other predictor. At the sample level, it was observed that for each additional hour of sleep, the slope of stress on emotion regulation became 0.038 points more negative, and vice versa. However, this interaction was not statistically significant, b= -0.038, SE= 0.11, t= -0.34, p= 0.739>.05. Therefore, there was insufficient evidence to state that the effect of either stress level or sleep duration on emotion-regulation performance was dependent on the level of the other predictor.

Figure 2.2

Fitted Multiple Linear Regression Plane for Sleep Duration, Stress Level, and Emotion Regulation



Note. The plane represents the multiple linear regression model. The X-dimension represents mean-centered sleep hours, the Y-dimension represents mean-centered stress level, and the Z-dimension represents the emotion-regulation-score outcome variable.

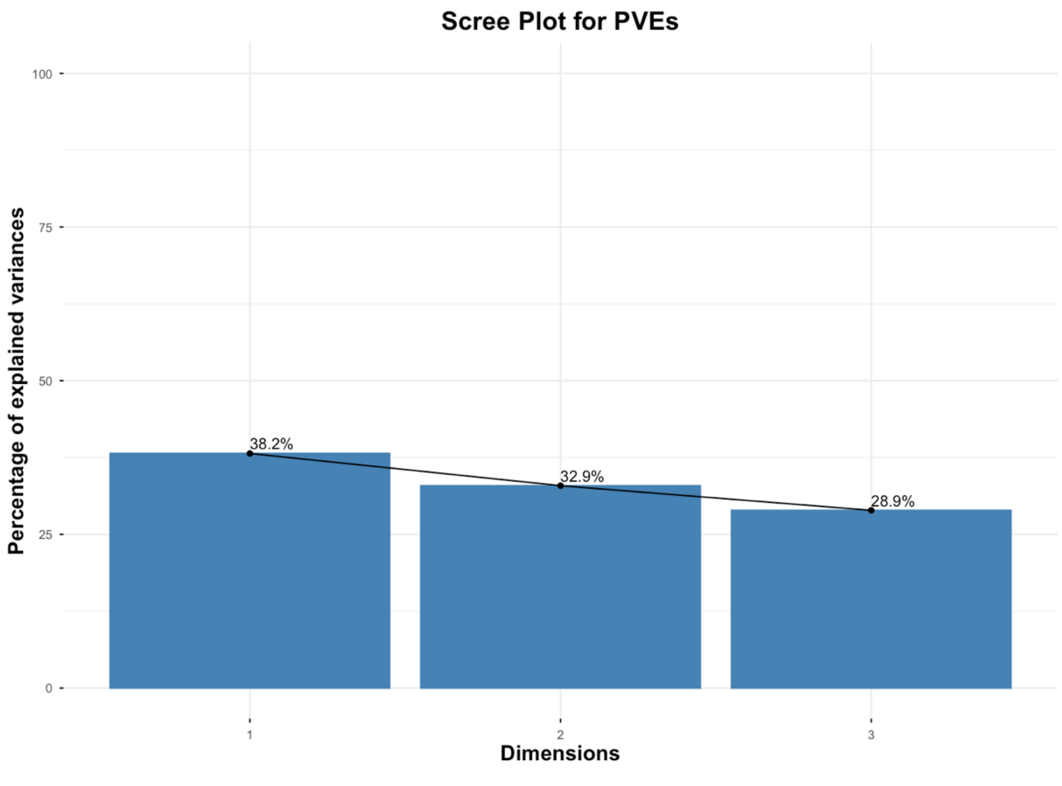
-AIM 1 Results and Discussion

At the beginning**,** an exploratory PCA was applied to summarize the three cognitive-performance measures. Here, I tentatively considered the possibility that the three measures might reflect a broader common dimension of cognitive performance (e.g., potential general cognitive ability). If such a common dimension were meaningfully represented by the PCA solution, the resulting component scores could be interpreted as an operational summary of individuals' relative performance along that broader dimension.

Figure 2.3 shows the scree plot of proportional eigenvalues, representing the proportion of the total variance that each component can possibly capture. It was found that they had approximately the same proportional eigenvalue, meaning that there was no dominant component preserving most of the information. Moreover, to determine the number of retained components, I relied on a widely used practical retention rule suggesting that retained components should cumulatively explain approximately 70% of the total variance (Cangelosi & Goriely, 2007; Jolliffe & Cadima, 2016). However, the compression is not justified based on the observed proportion of the total variance that the components can explain. On the other hand, the exploratory varimax-rotated solution further provided complementary evidence against a meaningful common cognitive-performance dimension. Each cognitive measure loaded almost exclusively on a different rotated component, with Stroop RT, PVT RT, and N-back accuracy showing loadings of approximately .999, .998, and .999 on distinct components, respectively. This almost one-to-one correspondence between individual measures and components suggests that the three task measures retained largely distinct variance rather than clustering around a shared cognitive-performance dimension. In other words, an approximate loading of one for each variable on one component left little of that variable's variance to be represented by the other rotated components. Therefore, reducing the three measures to a single summary component would result in a substantial loss of task-specific information and was not considered meaningful for the subsequent analyses.

Figure 2.3

Scree Plot of Proportional variance Explained by Each Principal Component



Note. Bars represent the proportion of total variance explained by each principal component, while the connected points illustrate the corresponding proportional eigenvalues. PC1, PC2, and PC3 accounted for 38.2%, 32.9%, and 28.9% of the total variance, respectively, implying a relatively even distribution of explained variance across components.

Therefore, as aforementioned, unsupported variable summarization created the need for separate models to examine associations between predictor variables and outcome variables. Aim2_model1 established a simple linear regression to examine the effect of mean-centered emotion-regulation scores on Stroop task RT. At the sample level, the result indicates that each one-point increase in emotion regulation above the sample mean was associated with a 0.01-second increase in emotion regulation score, but this result was not statistically significant, b = 0.01, SE = 0.07, t-value = 1.51 and p-value = 0.14 > 0.05. The model merely captured approximately 3.78% of the variance in emotion-regulation scores, R^2 = 0.0378, apparently leaving most of the outcome variance unexplained by this model. Figure 2.4 shows the relationship between emotion regulation scores and Stroop task RT with a confidence interval band.

Assumption diagnostics were then conducted to assess whether the model conditions supporting valid statistical inference were reasonably satisfied. Pre-written diagnostic functions were applied sequentially, except for multicollinearity diagnostics, which were not required for this single-predictor model. For normality, the Q-Q plot and Shapiro-Wilk (S-W) test provided visual and statistical evidence of deviation from normality, p=.03 < 0.05 in the S-W test. Homoscedasticity, independence of residuals, linearity, and outliers were checked, and no substantial violations were detected (for diagnostics on other assumptions, see “Figure” section in the report file). To reduce reliance on the normality assumption for inference, a nonparametric bootstrap was used to approximate the sampling distribution of the coefficient. The bootstrapped 95% confidence interval (5000 resamples) was [-.004, .02], including 0. Therefore, the bootstrap analysis likewise provided insufficient evidence that emotion-regulation scores were not associated with Stroop task RT at the population level.

Figure 2.4

Relationship between Mean-Centered Emotion Regulation Scores and Stroop Task RT (sec)

Note. The shaded area represents the 95% confidence interval around the fitted regression line for the estimated mean outcome at corresponding values of the emotion regulation scores.

Aim2_model2 established a simple linear regression to examine the effect of mean-centered stress level on PVT RT. At the sample level, the result indicates that each one-point increase in stress level above the sample mean was associated with a 0.94-second decrease in PVT RT, but this result was not statistically significant, b = -.94, SE = 1.04, t-value = -.90 and p-value = 0.37 > 0.05. The model merely captured approximately 1.40% of the variance in emotion-regulation scores, R^2 = 0.014, apparently leaving most of the outcome variance unexplained by this model. Figure 2.5 shows the relationship between stress level scores and PVT RT with a confidence interval band.

Assumption diagnostics were then conducted to assess whether the model conditions supporting valid statistical inference were reasonably satisfied. Pre-written diagnostic functions were applied sequentially, except for multicollinearity diagnostics, which were not required for this single-predictor model. For normality, the Q-Q plot and Shapiro-Wilk (S-W) test provided visual and statistical evidence of deviation from normality, p=.038 < 0.05 in the S-W test. Homoscedasticity, independence of residuals, linearity, and outliers were checked, and no substantial violations were detected (for diagnostics on other assumptions, see "Figure" section in the report file). To reduce reliance on the normality assumption for inference, a nonparametric bootstrap was used to approximate the sampling distribution of the coefficient. The bootstrapped 95% confidence interval (5000 resamples) was [-2.92, 1.15], including 0, and it was relatively wide, implying high uncertainty in the sample statistic for the effect of stress level on PVT RT performance. Therefore, the bootstrap analysis likewise provided insufficient evidence that stress level was not associated with PVT RT at the population level.

Figure 2.5

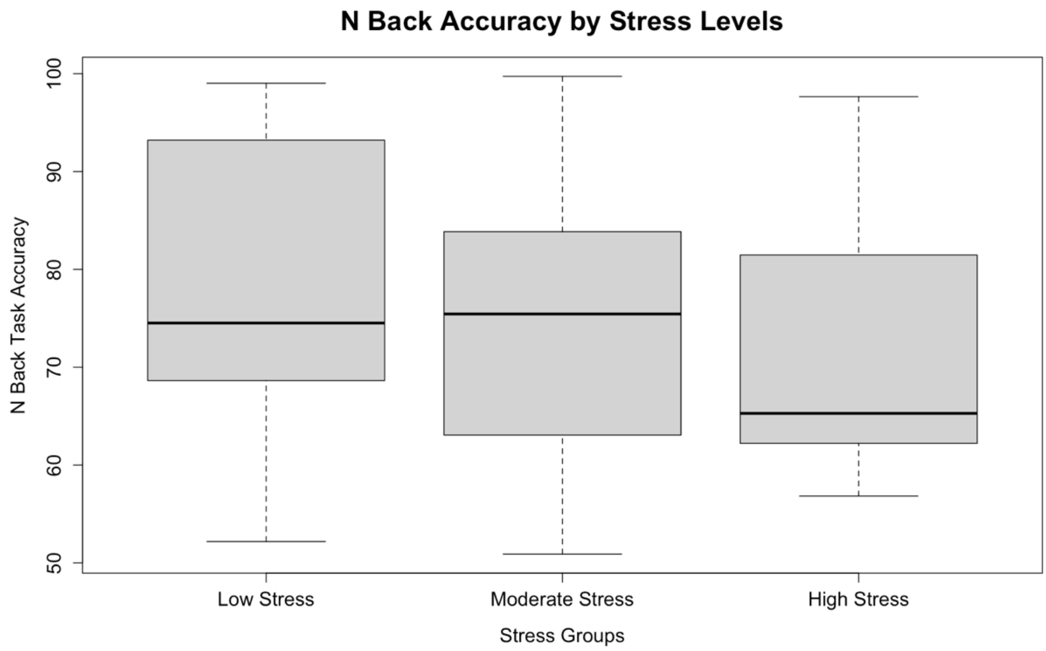
Relationship between Mean-Centered Stress Level Scores and PVT RT (sec)

Note. The shaded area represents the 95% confidence interval around the fitted regression line for the estimated mean outcome at corresponding values of the stress level scores.

The whole Aim2_model3, in essence, was established because I simply strived to examine the association between an individual's stress level and N-back accuracy from another, more intuitive perspective, considering that we generally tend to describe stress levels qualitatively. Aim2_model3 was therefore established to examine whether differences among low-, moderate-, and high-stress groups could meaningfully account for individual variation in N-back accuracy. Reduced_Aim2_model3 first constructed an ANOVA model to examine the unadjusted differences in N-back accuracy across the three stress groups, and the boxplot in Figure 2.6 provides a basic descriptive summary of N-back accuracy across these groups. The boxplot provides a descriptive comparison of N-back accuracy performance across the separate stress groups. It was observed that the medians of the low- and moderate-stress groups were similar, but the median for the high-stress group was relatively low. Considerable within-group variability was observed for each group, but there were no obvious outliers. The one-way ANOVA showed that differences in N-back accuracy performance were not statistically significant across the three stress groups, $F(2, 57) = 1.114$, $p = 0.335 > 0.05$. In practical terms, this means that there is insufficient evidence to support the conclusion that different stress groups produce differences in N-back accuracy performance.

Figure 2.6

Distribution of N-Back Accuracy Across Stress Groups (Boxplot)



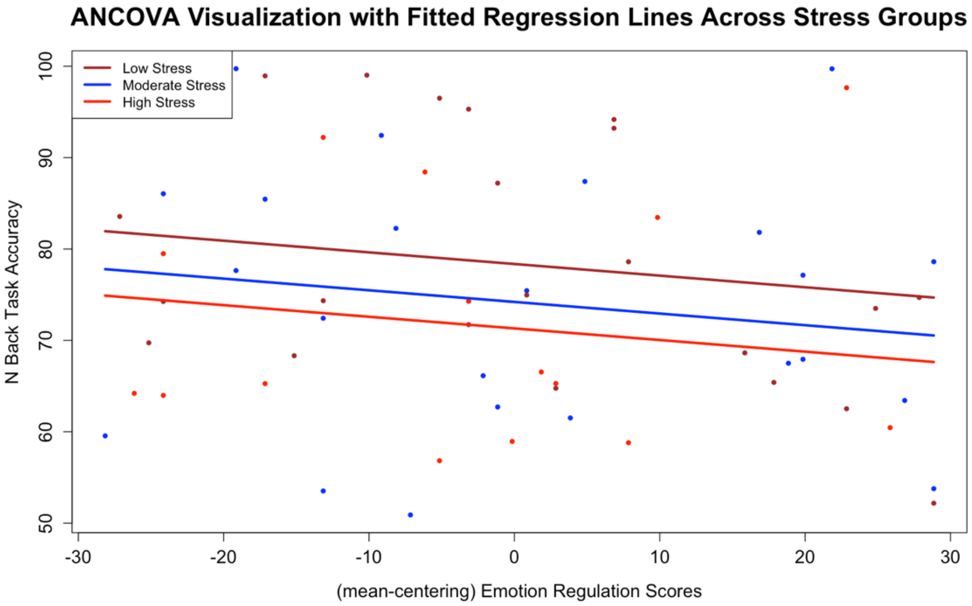
Note. The center line within each box represents the median N-back accuracy, the box represents the interquartile range (IQR), and the whiskers indicate the range of non-outlying observations within 1.5 × IQR.

Later, assumption diagnostics were conducted to assess whether the conditions supporting valid inference from the F-test were reasonably satisfied. Considering that ANOVA can be expressed as a special case of regression analysis, I directly applied the pre-written functions for regression assumption diagnostics. However, for homogeneity-of-variance testing, I used the function that is specifically designed for testing in ANOVA when the independent variable is categorical (see levenTest() function). It was found that the assumptions of normality of model errors and homogeneity of variance were approximately met, and no obvious outlier was detected. For homogeneity of variance, the p-value for Levene's test is approximately $0.85 > 0.05$, suggesting that there is insufficient evidence of a violation of homogeneity of variance. However, one discrepancy occurred for normality of errors. The Q-Q plot suggested some visual deviation from normality, whereas the Shapiro-Wilk test narrowly failed to reject the null hypothesis of normality, P-value $=.051 > 0.05$. Evidence regarding error normality was thus somewhat mixed, although there was insufficient statistical evidence to conclude that the normality assumption was violated.

Full_Aim2_model3 then constructed an ANCOVA by incorporating mean-centered emotion-regulation performance as a covariate into the established ANOVA model. Before conducting the formal analysis, the interaction between stress group and the covariate was tested to examine whether the association between emotion-regulation scores and N-back accuracy remained approximately constant across stress groups, as required by the homogeneity-of-regression-slopes assumption. The interaction term was not statistically significant, $p= .34> .05$, suggesting insufficient evidence of a violation of this ANCOVA assumption. After controlling for emotion-regulation performance, the one-way ANCOVA still showed a statistically non-significant effect of stress group on N-back accuracy, $F(2,56) =1.124$, $p=.332 >.05$. Specifically, this means that there was insufficient evidence that N-back accuracy differed across stress groups after statistically controlling for between-individual variation associated with emotion-regulation performance. The result indirectly suggested that individual differences in emotion-regulation performance did not account for a substantial amount of variation in N-back accuracy in the present sample. If the covariate had explained a substantial portion of the between-individual variation previously contained in the residual variance, controlling for emotion-regulation performance could have reduced the residual variance and consequently increased the F-value for the stress-group effect. However, only a minimal change in the F-value was observed ($dF \approx 0.01$). Figure 2.7 shows the relationships between distinct stress groups and N-back accuracy performance scores across emotion regulation scores at the sample level.

Figure 2.7

ANCOVA Visualization with Fitted Regression Lines Across Stress Groups



Note. Points represent observed N-back accuracy scores as a function of mean-centered emotion-regulation scores within distinct stress groups. Lines represent the fitted ANCOVA regression relationships for the low-, moderate-, and high-stress groups. The approximately parallel slopes are consistent with the absence of a statistically significant interaction term.

-OVERALL CONCLUSION

This data analysis project provided several intriguing findings that differed from the initial theoretical expectations. First, in the multivariate EDA module, the correlation-matrix heatmap indicated that none of the examined pairwise associations reached statistical significance in the present sample. For the specific examination of experience-related and psychophysiological-state predictors of cognitive performance, aim 1 provided insufficient evidence of a meaningful association between sleep duration and emotion-regulation performance. This finding did not support the initial theoretical expectation of a positive association; descriptively, the observed sample coefficient was even in the opposite direction. Subsequently, after stress level was incorporated as an additional predictor, multiple linear regression showed that neither the conditional association of sleep duration nor that of stress level with emotion regulation was statistically significant. The addition of stress also provided little incremental explanatory value beyond sleep duration alone, suggesting that stress level accounted for little additional variation in emotion-regulation performance in the present sample. Finally, the non-significant interaction term provided insufficient evidence that the association of either sleep duration or stress level with emotion regulation depended on the level of the other predictor. Therefore, the proposed buffering pattern, whereby greater sleep duration might attenuate the negative association between stress and emotion regulation, was not supported in the present sample. At the beginning of Aim 2, I attempted to summarize the three primary cognitive-performance measures into a principal component representing a potential common cognitive-performance dimension. Nevertheless, the PCA indicated that reducing the three measures to a single component would involve a substantial loss of information, or total variance. Consequently, the subsequent analyses treated the three cognitive-performance constructs separately. There was insufficient evidence of a statistically meaningful association between emotion-regulation performance and selective-attention performance on the Stroop task. Similarly, stress level was not significantly associated with sustained-attention and behavioral-alertness performance on the PVT task in the present dataset. Finally, there was insufficient evidence that N-back working-memory performance differed meaningfully across stress groups after controlling for emotion-regulation performance. The minimal change from the unadjusted ANOVA to the ANCOVA further suggested that emotion regulation accounted for relatively little additional variation in N-back performance beyond the stress-group model.

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In writing this report, I primarily used ChatGPT-4o, developed by OpenAI, to polish the manuscript's grammar, syntax, and overall readability. The primary idea, the logic of modeling, and the organization of interpretations are all guaranteed to be original.

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